Chem SBQ3: Buffers & Titrations

Ch 16 Aqueous Ionic Equilibrium · Summary + flashcards from your study session

Read tonight.
Flashcards at breakfast.

> **Teacher's exam note (p.8):** you only **calculate the initial pH** of a buffer. When acid or base is added, **describe** what happens (Le Chatelier / H-H) with **no calculation**. Always write the **"because"**.

1. Strong vs weak (know this first)

	Strong (100% breaks apart)	Weak (only a little)
Acids	HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$	HF, HNO_2, CH_3COOH, HCN, HClO, $HClO_2$, $HCHO_2$
Bases	NaOH, KOH	**NH_3**

Periodic table tricks:
• Group 17: **HCl, HBr, HI strong**, but **HF is weak** (F holds its H too tightly).
• Oxyacids: count **O − H**. If it is **2 or more**, the acid is strong (HNO_3 = 2, H_2SO_4 = 2, $HClO_4$ = 3). If it is 1 or 0, it is weak (HNO_2, $HClO_2$, HClO).

2. What is a buffer? (p.3–4)

A solution that **resists pH change** when a small amount of acid or base is added.
Weak acid + its conjugate base (acidic buffer, pH < 7) or **weak base + its conjugate acid** (basic buffer, pH > 7).
Swap trick: swap the H for Na to get the salt. HF → **NaF**, HNO_2 → **$NaNO_2$**, HCN → **NaCN**.

Acidic buffer	Basic buffer
HF / NaF, CH_3COOH / CH_3COONa, HNO_2 / $NaNO_2$	**NH_3 / NH_4Cl** (if you see NH_3, think basic buffer)
Add H^+: **$A^- + H^+ \rightarrow HA$** Add OH^-: **$HA + OH^- \rightarrow A^- + H_2O$**	Add H^+: **$NH_3 + H^+ \rightarrow NH_4^+$** Add OH^-: **$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$**

> ⚠ **Not a buffer:** anything with a **strong** acid (HCl + NaCl, HNO_3 + $NaNO_3$, HBr + NaBr), or a weak acid with the **wrong** salt (HNO_2 + NaCl).

3. Henderson-Hasselbalch (p.7–8, 11)

$$pH = pKa + \log\left(\frac{[base]}{[acid]}\right)$$

Ratio [base]/[acid]	log	pH	Example (pKa 4.74)
10	+1	pKa + 1	1.0 M base / 0.10 M acid → **5.74**
1 (equal)	**0**	**pH = pKa**	0.20 / 0.20 → **4.74**
0.1	−1	pKa − 1	0.10 M base / 1.0 M acid → **3.74**

Working order: 1) divide base ÷ acid, 2) take **log** (not ln), 3) add to pKa.
Backwards (find the ratio): log(ratio) = pH − pKa, so ratio = **10^(pH − pKa)**. Example: 4.25 − 3.74 = 0.51, so ratio = 10^0.51 = **3.24**.
pKa = −log Ka. Also: Ka × Kb = Kw, and pKa + pKb = 14.

4. Buffer effectiveness, capacity, range (p.9–11)

- **pH depends on the RATIO.** Same ratio gives the same pH, so **diluting doesn't change the pH**.
- **Capacity depends on CONCENTRATION.** More concentrated means it can absorb more acid or base.
- **Best buffer:** ratio = 1, so **pH = pKa**.
- **Buffer range = pKa ± 1** (ratio between 0.1 and 10).
- **Choosing an acid:** pick the one whose **pKa is closest** to the pH you want.

5. Common ion effect (p.1–2)

Add NaF to HF: the extra F^- shifts $HF \rightleftharpoons H^+ + F^-$ **LEFT**, so **fewer** HF break apart, so **% ionization goes DOWN**, **$[H^+]$ goes down**, and **pH goes UP**.
Data: HF alone has pH 1.57 and α 2.7%. HF + NaF has pH **3.14** and α **0.072%**.
% ionization = $[H^+] / C_{original} \times 100$.

⚠ pH works backwards: **less H^+ means a HIGHER pH**, and more H^+ means a lower pH.

6. Titration (p.13–25)

Titrant: known concentration, in the burette. **Analyte**: unknown concentration, in the flask.
Equivalence point: moles of acid = moles of base (set by stoichiometry).
End point: where the **indicator changes colour**.

Titration	Curve starts	pH at equivalence	Why
Strong acid + strong base	Low (≈1)	**= 7**	Only a neutral salt is left (NaCl)
Weak acid + strong base	≈3, small initial rise	**> 7**	The conjugate **base** (e.g. CH_3COO^-) is left
Weak base + strong acid	High, then goes down	**< 7**	The conjugate **acid** (NH_4^+) is left

Trick: the pH at equivalence goes toward whatever is **strong**.
Half-equivalence point: pH = pKa, because [acid] = [base] there. This is the best buffering region.

Flashcards (fold the page or cover the bottom half)

> Yellow cards = **your mistakes from today**. Do those first.

Which one is a basic buffer?

NH_3 + NH_4Cl (weak base + its salt). pH > 7.

The 6 strong acids?

HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$. **HF is weak!**

Half-equivalence point: pH = ?

pH = pKa

Less H^+ means pH goes…?

UP (less acidic)

Weak base + strong acid: pH at equivalence?

< 7 (NH_4^+ left over is an acid)

Conjugate base of HNO_2?

NO_2^-. Don't forget the negative charge!

H-H equation

pH = pKa + log([base]/[acid])

[base] = [acid], so pH = ?

pH = pKa (log 1 = 0)

Buffer range?

pKa ± 1

pH depends on…? Capacity depends on…?

pH: the **ratio**. Capacity: the **concentration**.

Add NaF to HF: shift / α / pH?

Left / down / up

Strong + strong: pH at equivalence?

= 7